Presentation 2

Link to presentation PDF, and code:

- Link to presentation:

  https://docs.google.com/presentation/d/1N8WhefrT7KPJaW6wIBDIsuflpjJDXcJsr7ZxrrPKVpI/edit?usp=sharing

- Link to github repo:

  https://github.com/arvindkrishna87/Team1_FullStack/tree/main/Presentation2

  - Grace's work: https://github.com/arvindkrishna87/Team1_FullStack/tree/main/Presentation2/Code1_HierarchicalMIL_06Nov_team
  - Nolan's work: https://github.com/arvindkrishna87/Team1_FullStack/commit/a6456243ff82b42e1e63dab0adb02787c434aa26 https://github.com/arvindkrishna87/Team1_FullStack/commit/dbab90f1b06efa33cf5520b159162ee3bbde1fdb
  - Junho's work: https://github.com/arvindkrishna87/Team1_FullStack/tree/main/Presentation2/Code2_NestedCVDesign_06Nov_Junho

What did each of the team members do?

Grace worked on analyst choice 1, pt. 1, which was to pair each patch with its score by name. In order to do so, Grace fixed mismatches between the patch filenames, embeddings, and attention weights by tracking the filenames throughout the pipeline. Moreover, Grace removed fallbacks and silent load failures to raise explicit errors. Finally, Grace developed and ran two tests to confirm that the filenames were correctly matched after shuffling and within the attention analysis.

Junho worked as the Mathlete and designed the Optuna search that nested cross-validation runs in each outer fold. He considered eighteen hyperparameters and chose five to tune, the learning rate, weight decay, classifier dropout, the attention-entropy weight and projector dropout, with a stated reason for fixing each of the other thirteen. He derived each search scale from how the hyperparameter acts in training and found that the previous weight-decay range shrinks the weights by at most 0.34% over a run, so he re-parameterised it as the total decay reached by epoch 4. He showed that at the loss minimum a class weight only rescales the predicted odds, so its best value against plain cross-entropy is 1, and he fixed it at the class ratio of about

1.67, which makes the training loss match balanced cross-entropy, the search objective. Pilot runs of one inner model took 10 to 46 minutes and showed that two trials with the same hyperparameters still differ by about 0.05 from the seed alone, so trials closer than about 0.1 cannot be reliably ranked. This set a budget of 40 trials with four inner models per outer fold, or 800 runs. The search starts with 16 scrambled Sobol points and continues with multivariate TPE whose good set holds at least six trials, in a single stage without pruning.

Sarah and Ashley worked as Consultants. They looked into how UNI2 was trained as a model, observed how it differed to KimiaNet, and evaluated how these differences may affect performance for our project. Ashley explored the implications of different training image sizes, with KimiaNet being trained on 1000x1000 images and UNI2 trained on 224x224 images. Sarah explored the implications of KimiaNet treating varying patch sizes within the model versus UNI2 resizing and center cropping prior to the model.

Nolan worked as the Analyst, extending the professor's SIFT-based patch finder (find_patch.py) into a new tool, top_patches_overlay.py, that reads a model run's attention output and outlines the top five attention patches on each full-resolution slice, color-coded by rank. He ran it on two slices per stain, Melan-A (cases 83 and 86), H&E (cases 3 and 86) and Sox10 (cases 34 and 90), and presented the overlays alongside patch-by-patch match checks and observations on where the model focuses its attention. He also discussed what has worked well and what could be improved in the coming weeks.

How does it help the project?

Grace’s work helps the project by improving the accuracy and interpretability of the model’s attention analysis. This part of the pipeline is critical for the stakeholders to understand what is driving the model’s decisions to classify a case as high-grade or benign. This includes knowing which patches are contributing the most to the decision and being able to visualize them. Explicit error handling also prevents missing or unreadable data from being ignored, which is important for the stakeholders if they add more data and need to understand what is going wrong.

Junho's design lets the team tune the model without letting the test fold influence the hyperparameters, so the accuracy reported after tuning is not inflated by tuning on test cases. It spends the search only on settings that change training, fixes the class weight that the previous search tuned without effect, and moves weight decay into a range where it acts. It also sizes the search to the noise measured in pilot runs, so the search fits the cluster budget and the chosen settings are less likely to come from seed luck.

The summary table and sampler settings can be implemented directly in the Optuna script.

Sarah and Ashley's work helps the project by identifying strengths and weaknesses of KimiaNet as compared to UNI2. This provided insights as to whether adopting UNI2 could be beneficial and improve model performance. Their work highlighted areas in which UNI2 could present itself as a stronger model for our project, while also calling out considerations and potential trade-offs, such as UNI2's relative weakness in identifying spatial relationships as compared to KimiaNet.

Nolan's work helps improve visualizations and align the model's decisions with the underlying pathology by confirming showing the most influential patches in the context of their original patient slices.

Issues faced that haven't been resolved:

For Grace's work, there are no big issues that have not been resolved for this analyst choice 1, pt. 1 task. Note there is still one silent load for patches smaller than 32x32 that do not have embeddings. There could be a way to mark these cases so that the stakeholders know what they are.

Junho: The design has not been run yet. The noise floor comes from one validation fold and four seeds, so it is approximate, with a wide 95% interval for the seed spread (0.04 to 0.26). The decay parameter is defined at epoch 4, the pilot's best epoch, which held when the learning rate was lowered fourfold; at a learning rate of 10^{-3} the best epoch was 11, so the decay reached there is larger than D states.

Nolan identified color differences between patches and slices. His script also isn't generalized for every case and requires manual .tif uploads. Finally, the top 5% patch .csv it relies on does not ensure that every slice has a full 5 patches visualized

Proposed next steps:

One proposed next step for Grace's work is to expand testing across the pipeline and by using the actual data rather than the artificially created data. It may also be useful to go through the attention analysis code and possibly re-evaluate and further optimize the attention analysis code.

Junho proposes running outer fold A (160 runs) to check where the selected values land before running the other four folds.

Nolan: Generalize the new script to work for all cases and slices without manual input.

References

Grace built on the methodology that has been created by STAT 390 students over the past two years and the updates that were made after presentation 1. This includes the data preprocessing and architecture exploration done in the 2024-25 year and model development, training, and testing done in the 2025-26 year. Moreover, Grace used the help of AI (ChatGPT) to help build the two test checks that were used. Grace prompted the AI with the test case ideas she had, and the AI helped to write out the code.

Junho built on the Presentation 2 base code and objectives from the STAT390 instruction team and on Team 3's Presentation 1 list of hyperparameters. Literature used includes Kingma and Ba (2015) on Adam's step size, Loshchilov and Hutter (2019) on decoupled weight decay, Gneiting and Raftery (2007) on proper scoring rules, Bergstra et al. (2011) on TPE, Falkner et al. (2018) on BOHB, and Owen (1998) on scrambled Sobol points. Claude (Anthropic) helped read the code and run the checks.

Sarah and Ashley built on literature journal, documentation, and code for UNI2 implemented by the coders as well as previously used code for KimiaNet. They also used outside expert sources and research done with ViT models and documentation to further understand code functions.

Nolan built on the presentation 2 base code, adding a new script top_patches_overlay.py that he ran on outputs from our recent run data folders within /projects/e32998/MIL_training/final_runs/run_20261103_164248. The new script incorporated Prof. Krishna’s find_patch.py file.